Lecture MT2 review

Tuesday, 18 April 2023 12:09 PM

You only always know the local reaction rate (rate assuming you know the local concentration)

} Idea of effectiveness factor ↓ to get actual reaction rate

(η corrects/accounts for transport effects)

Coarse grain model (Zurkely?)

Zukaly

η → changes along the axial reactor

$$D\frac{d^2c}{dx^2} - V\frac{dc}{dx} - k'C = 0$$

Optimization of Chemical Reactors

Optimality Principle

Fixing output conc. & minimizing the residence time